

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

2nd Semester of B.Tech.(CL/EC/EE/ME/IT/CE) University Examination May -2014

MA 102 : Engineering Mathematics – II

Date: - 12/05/2014(Monday)

Time: 01:30 p.m. to 04:30 p.m.

Total marks : 70

Instruction :

- 1) The question paper comprises of two sections.
- 2) Section I and Section II must be attempted in separate answer sheets.
- 3) Make suitable assumptions and draw neat figures wherever required.
- 4) Use of scientific calculator is allowed.

Section - I

Q.1 Do as directed.

- (a) Obtain 1st order ordinary differential equation satisfied by $y = mx$ (01)
- (b) Show that $(x^2 + y^2)dx + 2xydy = 0$ is exact differential equation and find solution. (02)
- (c) Find partial differential equation satisfied by $z = ax + by + ab$ (02)
- (d) Find general solution of 2nd order ordinary differential equation $y'' + 5y' + 6y = 0$ (02)

Q.2 Do as directed.

- (a) Obtain particular solution of $\frac{dy}{dx} - 2yx = y^2x$ with $y(0)=1$ (04)
- (b) Find general solution of $(D^2 - 2D + 5)y = 5x^3 - 6x^2 + 6x$ by using method of undetermined coefficients. (05)
- (c) Find solution of Lagrange's linear partial differential equation $xp + yq = z$ (05)

OR

Q.2 Do as directed.

- (a) Show that the differential equation $(1+xy)y dx + (1-xy)x dy = 0$ is non exact and hence find general solution of it. (04)
- (b) Find general solution of $y'' + 4y = \tan 2x$ differential equation by using the method of variation of parameters. (05)
- (c) Find Solution of linear partial differential equation $(y^2 + z^2)p - xyq = xz$ (05)

Q.3 Do as directed.

- (a) Find solution of 1st order ordinary linear differential $\frac{dy}{dx} + 2xy = 2e^{-x^2}$ (04)
- (b) Find general solution of $y'' - 4y = \sin 2x \cos 2x$ differential equation by using the method of undetermined coefficients. (05)
- (c) Identify types of $\sqrt{p} + \sqrt{q} = 1$ and $z = xp + yq + (p+q)^2$ non linear partial differential equation and hence find solution. (05)

OR

Q.3 Do as directed.

- (a) Find solution of $\frac{dy}{dx} = \frac{y^3 + x^3}{xy^2}$ (04)
- (b) Find general solution of $y''' - 6y'' + 11y' - 6y = 0$ (05)
- (c) Find solution of following Lagrange's linear partial differential equation (05)
$$x^2(y-z)p + y^2(z-x)q = z^2(x-y)$$

Section - II

Q.4 Do as directed.

(a) Show that $\int_a^\infty e^{-(2x-a)^2} dx = \frac{\sqrt{\pi}}{4} (1 - \operatorname{erf}(a))$ (02)

(b) Evaluate $I = \int_1^2 \int_3^4 dx dy$ (02)

(c) Find Eigen value of $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$ (03)

Q.5 Do as directed.

(a) Prove that $J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$ (04)

(b) Change the order of integration in $\int_0^1 \int_0^{1-x} x dy dx$ and hence evaluate it. (05)

(c) Obtain Eigen values and Eigen vectors of $A = \begin{bmatrix} 1 & 1 & 4 \\ 0 & 3 & 6 \\ 0 & 0 & -1 \end{bmatrix}$ (05)

OR

Q.5 Do as directed.

(a) Derive values of $P_3(x)$ and $P_1(x)$, using it show that $x^3 = \left(\frac{2}{5}\right)P_3(x) + \left(\frac{3}{5}\right)P_1(x)$ (04)

(b) Find volume of a cube bounded by $0 \leq x \leq 1$, $0 \leq y \leq 2$ & $0 \leq z \leq 3$ (05)

(c) Verify Cayley - Hamilton theorem for $A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$ and hence prove that (05)

$$A^5 - 3A^4 + A^3 - 7A^2 + 5A + I_2 = 61I_2 - 43A$$

Q.6 Do as directed.

(a) Show that $\int_0^1 \frac{1}{\sqrt{1-x^4}} dx = \frac{\sqrt{\pi} \Gamma(1/4)}{4\Gamma(3/4)}$ (04)

(b) Find volume bounded by cylinder $x^2 + y^2 = 4$ between the planes $y + z = 3$ and $z = 0$ (05)

(c) If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, then find A^2, A^3, A^{-1} & A^{-2} using Cayley - Hamilton theorem. (05)

OR

Q.6 Do as directed.

(a) Show that $J_{\frac{5}{2}}(x) = \sqrt{\frac{2}{x\pi}} \left[\left(\frac{3-x^2}{x^2} \right) \sin x - \frac{3}{x} \cos x \right]$ and $J_{\frac{3}{2}}(x) = \sqrt{\frac{2}{x\pi}} \left[\frac{\sin x}{x} - \cos x \right]$ (04)

(b) Evaluate $\int_0^\infty \int_0^\infty e^{-x^2-y^2} dy dx$ by changing it in to polar coordinates. (05)

(c) Find area bounded by $r = 1 + \sin \theta$ by using polar coordinates. (05)
